(7) Let $f(x) = \begin{cases} 0, & \text{if } x < 0 \\ x^2, & \text{if } 0 \le x < 2 \\ 4, & \text{if } x \ge 2 \end{cases}$ .Graph the function. Show that $f(x)$ continuous on $(-\infty, \infty)$.

(8) If f and g are continuous functions with $f(3) = 5$ and $\lim_{x \to 3} [2f(x) - g(x)] = 4$, find $g(3)$.

(9) Find the points at which f is discontinuous. At which of these points f is continuous from the right, from the left, or neither? Sketch the graph of f.

(i) $f(x) = \begin{cases} 2x+1, & \text{if } x \le -1 \\ 3x & \text{if } -1 < x < 1 \\ 2x-1, & \text{if } x \ge 1 \end{cases}$ (ii) $f(x) = \begin{cases} (x-1)^3, & \text{if } x < 0 \\ (x+1)^3, & \text{if } x \ge 0 \end{cases}$

(10) A function f is defined as follows :

$$f(x) = \begin{cases} 0 & \text{for } x < 0; \\ x & \text{for } 0 \le x < 1; \\ -x^2 + 4x - 2 & \text{for } 1 \le x < 3; \\ 4 - x & \text{for } x \ge 3 \end{cases}$$

Is the function continuous?

(11) Which of the following functions f has a removable discontinuity at $x = x_0$? If the discontinuity is removable, find a function g that agrees with f for $x \neq x_0$ and is continuous on $\mathbb{R}$.

(i) $f(x) = \dfrac{x^2 - 2x - 8}{x + 2}$, $x_0 = -2$.

(ii) $f(x) = \dfrac{x^3 + 64}{x + 4}$, $x_0 = -4$.

(iii) $f(x) = \dfrac{3 - \sqrt{x}}{9 - x}$, $x_0 = 9$.

(12) Find the constant b that makes g continuous on $(-\infty, \infty)$.

$$g(x) = \begin{cases} x^2 - b^2 & \text{if } x < 4 \\ bx + 20 & \text{if } x \ge 4 \end{cases}$$

(13) Consider the function $f(x) = x \sin \dfrac{\pi}{x}$. What value must we give $f(0)$ in order to make the function continuous everywhere?

(14) The function $f(x) = \dfrac{x^2 - 1}{x^3 - 1}$ is not defined at $x = 1$. What value must we give $f(1)$ inorder to make $f(x)$ continuous at $x = 1$?

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